Statics of Solid Mechanics

-ME 164-

Fundamental Concepts and Definitions

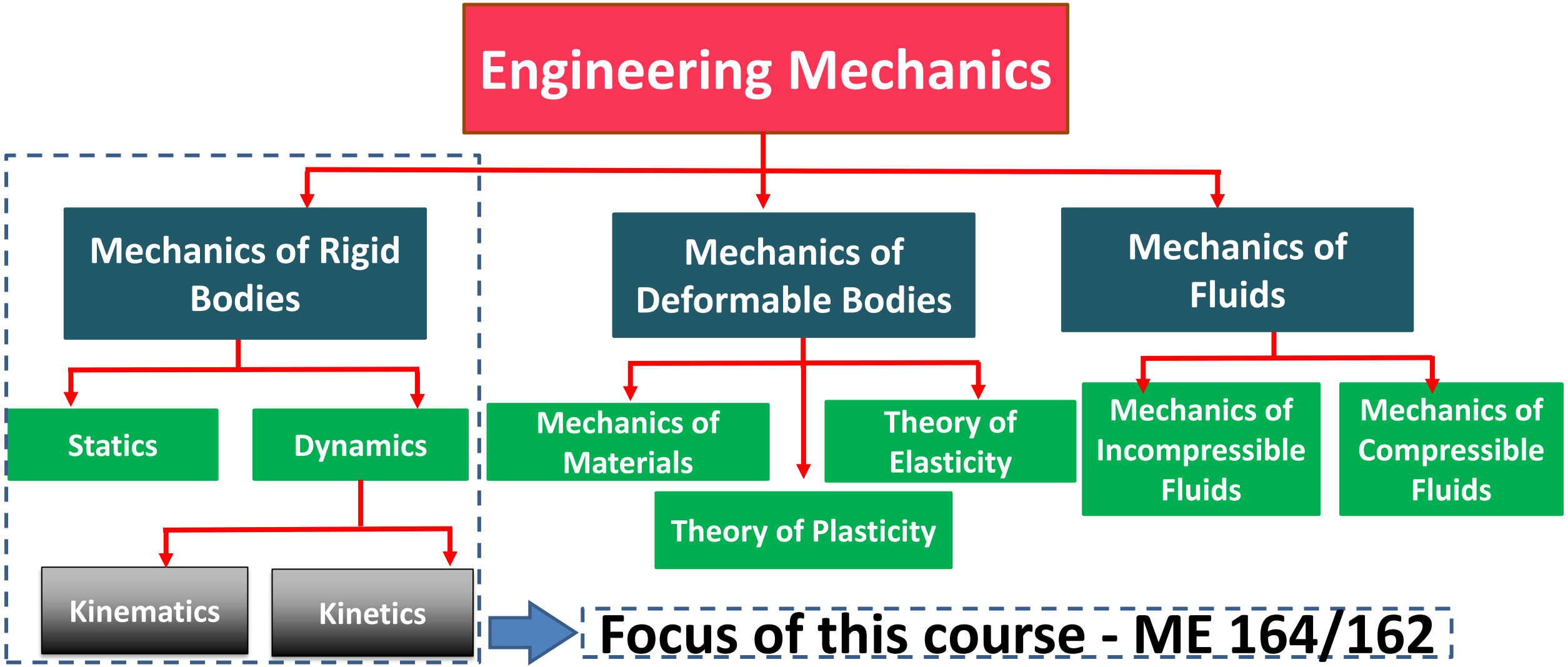
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About Engineering Mechanics

Mechanics is that branch of physical sciences that deals with the condition of rest or motion of bodies under the action of forces.

About Engineering Mechanics

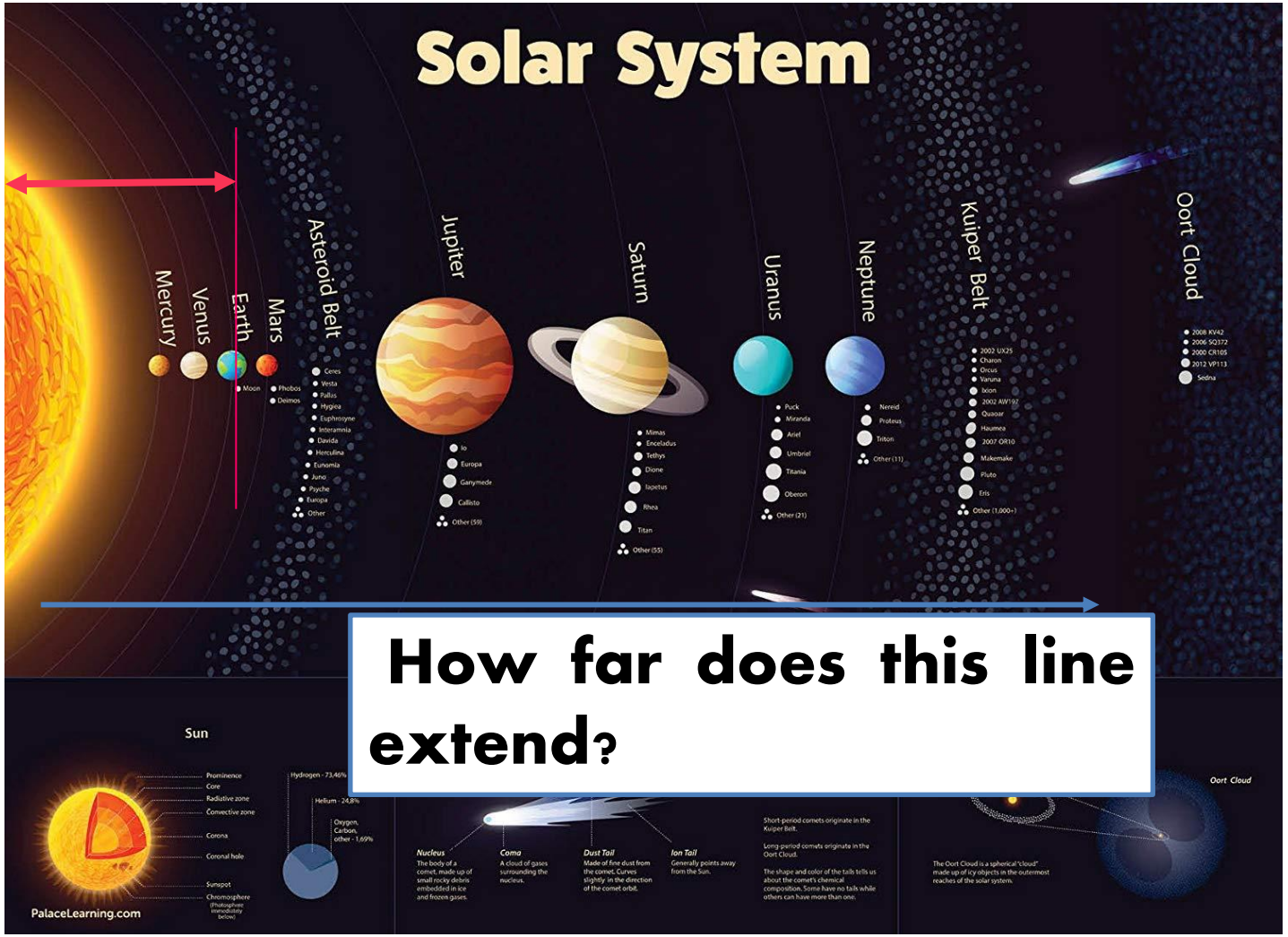


Fundamental Concepts and Definitions

- **Space and Time:**
  - **Space:** A collection of all positions in our universe that a point may occupy.
  - Because space is 3-dimensional, 3 pieces of information, called **Coordinates** are required to locate a point in space (more on this later).
  - **Time**: A measure of when an event or sequence of events occur.

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Fundamental Concepts and Definitions



Mean earth-sun distance:
– 150 million km

ACTIVITY

How long will it take for light to travel from the sun to the earth?

Fundamental Concepts and Definitions

- **Mass and Force**
  - **Mass** is the amount of matter, or material, in an object.
  - **Force** is an agency that is capable of producing motion of an object.

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Fundamental Concepts and Definitions

- A Force may be categorized as either **Contact Force** or **Field Force**.
- **Contact forces** arise when two bodies touch. The **contact forces** are distributed over a finite area of contact and have dimensions of force/area.
- **Contact forces** act in both normal and tangential directions. The tangential component is also called friction force.
- Examples of **contact forces** include forces between the feet and the ground when one is standing or forces acting on a building when wind is blowing.

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Fundamental Concepts and Definitions

- **Field Force is a force between bodies that acts through space. Field Forces (also called body forces) act throughout the volume of an object have the dimensions of force/volume.**

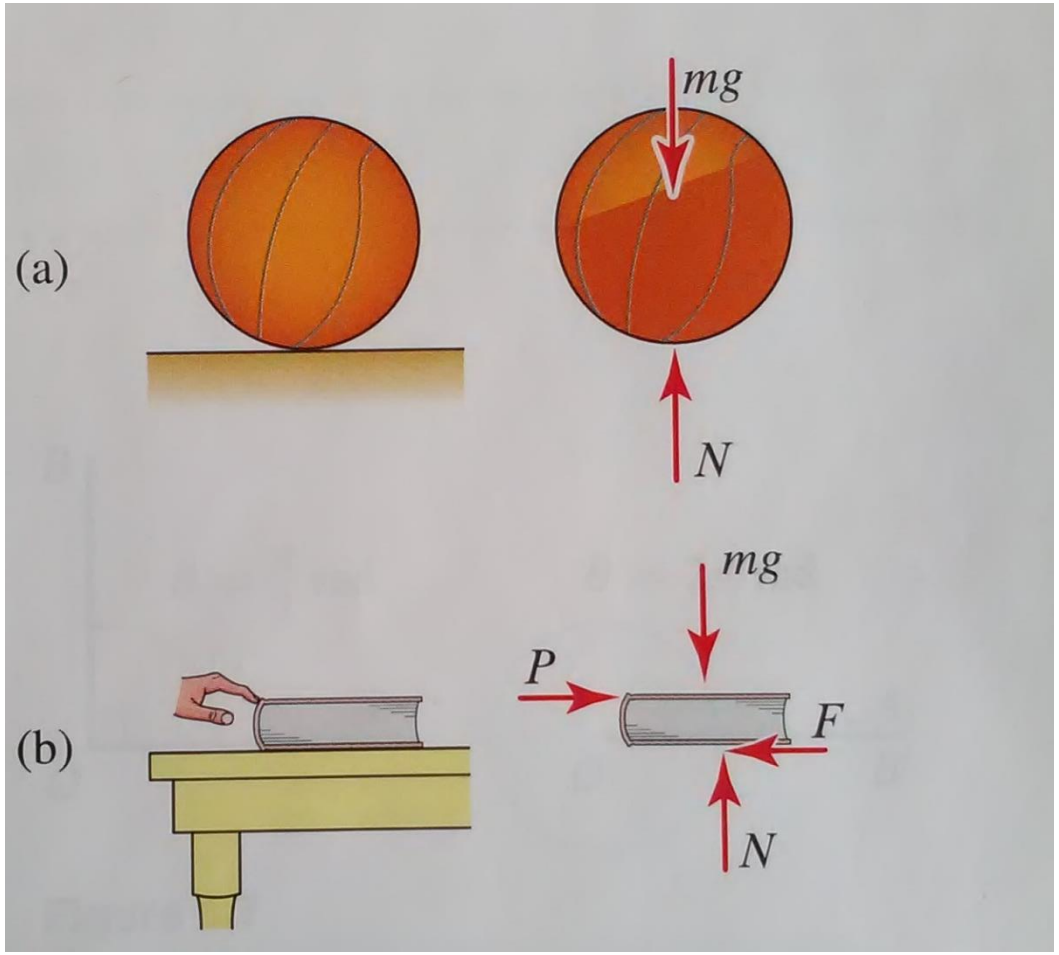
- **Examples of field forces include the weight of an object, the attractive force between the moon and earth, and force of attraction between a magnet and an iron.**

- **In many applications, a field force may be represented by a concentrated force that acts at a point.**

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Fundamental Concepts and Definitions

- **Examples of Field and Contact forces.**



In both cases, the field force is the weight (W=mg).

The Contact forces are the Normal Force N, the applied force P, and the friction force F.

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Fundamental Concepts and Definitions

- A **rigid body** is a body that is not **deformable** and hence the distance between points in the body never changes.
- A when a **deformable** body is subjected to forces, distances between points in the body may change.
- There are not true rigid bodies in nature, it is an **idealization** that provides considerable simplification during analysis.
- This course will focus on mechanics of **rigid bodies**.

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Fundamental Concepts and Definitions

- **A Scalar is a quantity that is completely characterized by a single number.**
- **Examples of scalars include: temperature, length, mass and density.**
- **A vector on the other hand has both size (or magnitude) and direction (more on vectors soon).**
- **Examples of vectors include: position (relative to an origin), velocity acceleration and force.**
- $\vec{v} = \frac{d\vec{r}}{dt}$; $\vec{a} = \frac{d\vec{v}}{dt}$

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Fundamental Concepts and Definitions

Newton's Laws of Motion

- **Law 1: A particle remains at rest or continues to move in a straight line with uniform velocity, if there is no unbalanced force acting on it.**

- **Law 2: The acceleration of a particle is proportional to the resultant force acting on the particle and is in the direction of this force.**

- **Mathematically:**
  - $\vec{F} = m\vec{a}$ **where** $\vec{F}$**, is the resultant force acting on the particle,** $\vec{a}$ **is the acceleration of the particle, and the constant of proportionality is the mass of the particle** m**.**

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Fundamental Concepts and Definitions

Newton's Laws of Motion

- **A more general form of the Second law is given by:**

$$\vec{F} = \frac{d(m\vec{v})}{dt}$$

- **Where $\vec{v}$ is the velocity of the particle and, $\frac{d(m\vec{v})}{dt}$ is the time rate of change of $m\vec{v}$ (called momentum).**

- **When mass is constant the expression reduces to $\vec{F} = m\vec{a}$ (a special case).**
- **The special case is not ($\vec{F} = m\vec{a}$) is not valid in applications in which mass is not constant (e.g. motion of a rocket).**

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Fundamental Concepts and Definitions

Newton's Laws of Motion

- **Law 3: The forces of action and reaction between interacting bodies are equal in magnitude, opposite in direction and collinear.**

- **Comments on Newton's Laws of Motion.**

  - **There is no fundamental proof of validity of the laws of motion. Their validity is proven by experiments and centuries of applications.**

  - **The First Law is, in fact, implied by the Second Law.**

  - **The second law is the most important fundamental principle upon which mechanics in general is based.**

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Units of measurement and dimensions

- **Dimensions are a measurable extent of some kind (e.g. length, mass, time).**
- **Units on the other hand are used to measure dimensions (e.g. meter, kilogram, second).**
- **Fundamental quantities cannot be defined in terms of other physical quantities. E.g time(T), length(L), mass(M)**
- **The SI system of units includes three classes of units: (1) base units, (2) supplementary units, and (3) derived units.**

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Units of measurement and dimensions

The SI is made up of seven (7) base units that define twenty-two (22) derived units with special names and symbols.

7 Base Units

Quantity	Name of Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric Current	Ampere	A
Temperature	Kelvin	K
Amount of Substance	mole	mol
Luminous intensity	candela	cd

Units of measurement and dimensions

Supplementary units:

Quantity	Name of Unit	Symbol
Plane Angle	radian	rad
Solid Angle	steradian	sr

Derived units are expressed algebraically in terms of base units and/or supplementary units.

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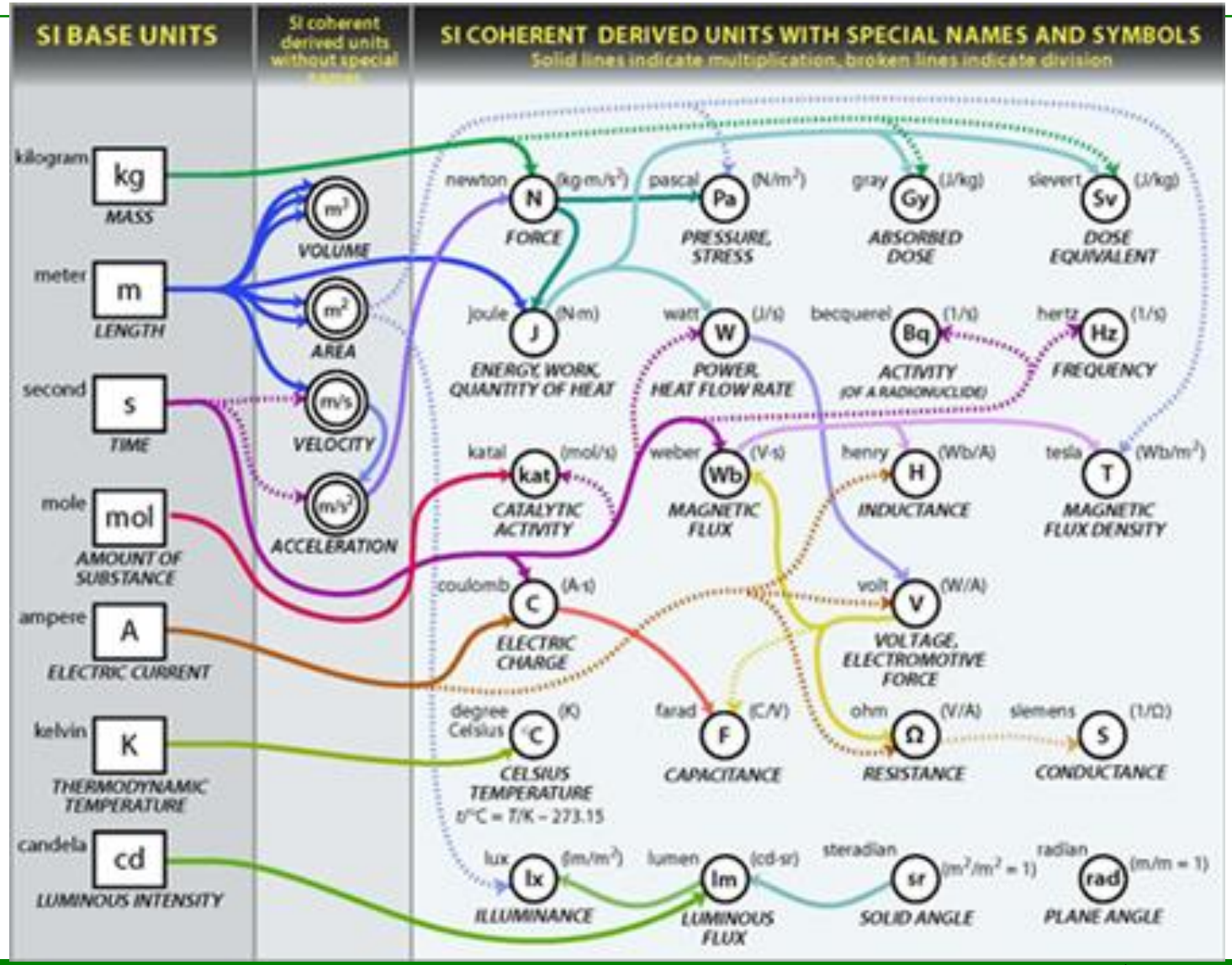
Units of measurement and dimensions

Derived units (examples):

Quantity	Derived Unit	Special Name	Symbol
Force	Kilogram meter per second squared ($kg.m/s^2$)	Newton	N
Pressure and Stress	Newton per meter squared (N/m^2)	Pascal	Pa
Work and Energy	Newton.meter (N.m)	Joule	J
Power	Joule per second (J/s)	Watt	W
Area	square meter	-	m^2
Density	Kilogram per cubic meter	-	kg/m^3

Units of measurement and dimensions

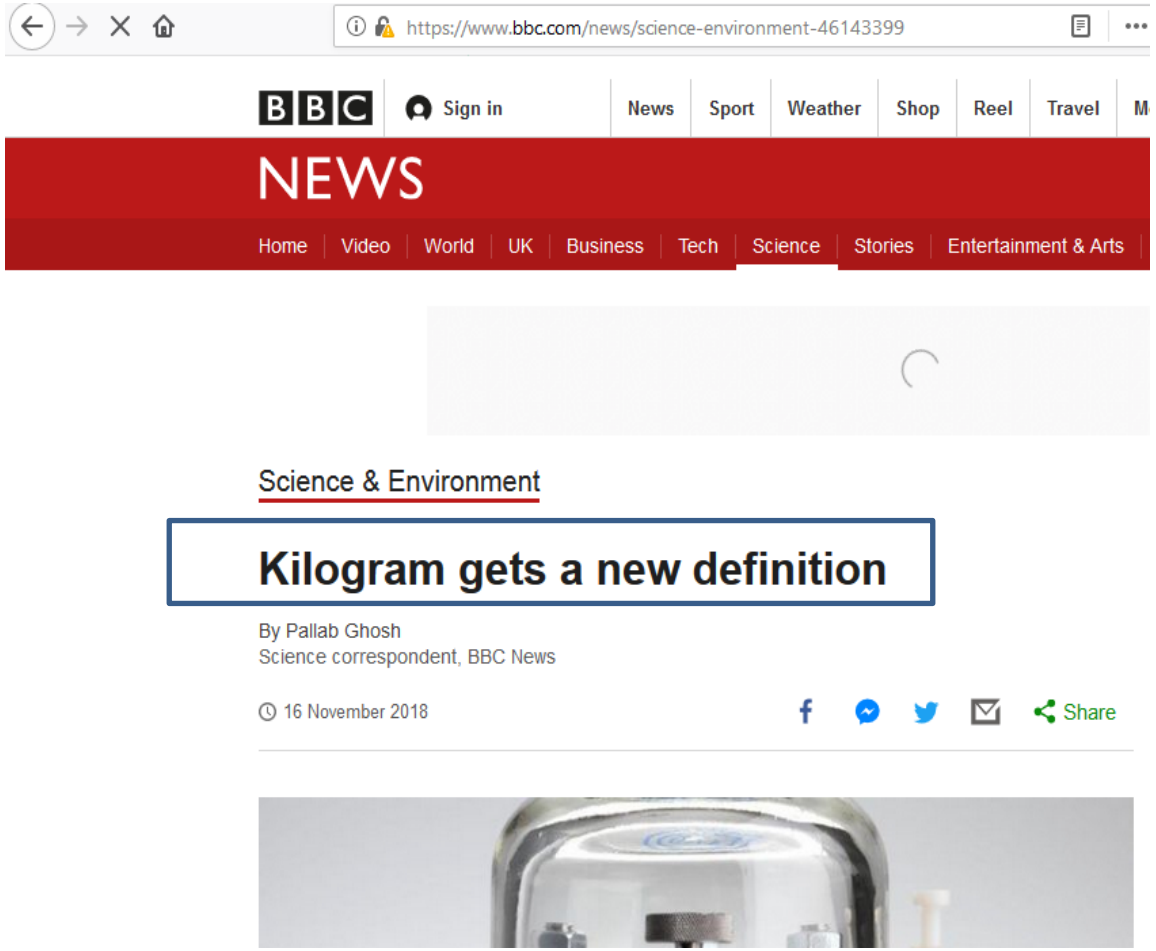
Relationship between base units and derived units.



Read more at: https://www.nist.gov/pml/weights-and-measures/metric-si/si-units

Units of measurement and dimensions

- **Fundamental units have very precise definitions.**
- **E.g. kilogram has been defined as the exact weight of a cylinder of platinum-iridium held in a locked vault in Paris.**
- **The meter is the length of the path travelled by light in vacuum during a time interval of 1/299 792 458 of a second.**



https://www.bbc.com/news/science-environment-46143399

Units of measurement and dimensions

Further reading:

- https://physics.nist.gov/cuu/Units/meter.html

Prefixes of SI Units

Prefix	Prefix Symbol	Word	Conventional Notation	Exponential Notation
tera	T	trillion	1,000,000,000,000	$1x10^{12}$
giga	G	billion	1,000,000,000	$1x10^{9}$
mega	M	million	1,000,000	$1x10^{6}$
kilo	k	thousand	1,000	$1x10^{3}$
hecto	h	hundred	100	$1x10^{2}$
deka	da	ten	10	$1x10^{1}$
-----	----	one	1	$1x10^{0}$
deci	d	tenth	0.1	$1x10^{-1}$
centi	c	hundredth	0.01	$1x10^{-2}$
milli	m	thousandth	0.001	$1x10^{-3}$
micro	μ	millionth	0.000001	$1x10^{-6}$
nano	n	billionth	0.000000001	$1x10^{-9}$
pico	p	trillionth	0.000000000001	$1x10^{-12}$
femto	f	quadrillionth	0.000000000000001	$1x10^{-15}$

It will be great to memorize these prefixes – at least the widely-used ones.

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Dimensional Homogeneity

Dimensional homogeneity is required for an equation to be valid; i.e. every term in an equation must have the same unit.

left-hand side = right-hand side

Symols of base dimensions:

Length = L, Time= T, Mass = M, etc.

E.g. the dimensions of density may therefore be expressed as:

$$ML^{-3}$$

Checking dimensions can serve as a valuable tool to spot errors.

Dimensional Homogeneity

Identities for unit conversion are an "exception" to this rule.

Examples include:

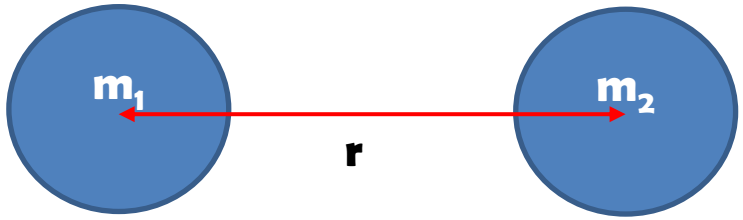
$$1000\ mm = 1\ m$$

$$1\ N\ = 1\ kg.m/s^2$$

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Law of Universal Gravitation

Every object in the Universe attracts every other object with a force directed along the line of centers for the two objects that is proportional to the product of their masses and inversely proportional to the square of the separation between the two objects.



$$F = G\frac{m_1 m_2}{r^2}$$

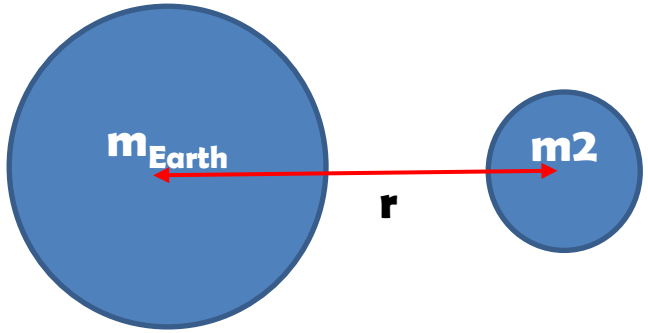
Where:

m_1,m_1 = masses of particles 1 and 2 (kg);
r = distance between the particles (m);
G= universal gravitational constant, approximately 6.673×10^{-11} $m^3/(kg.s^2)$
F = force of attraction between two particles (N)

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Law of Universal Gravitation

The earth's gravitational pull:



$$F = G\frac{m_{Earth} \times m_2}{r^2}$$

This force is called the Weight.

9.81 m/s²

Where:

m_{Earth} = Mass of the earth (5.972×10^{24} kg)
m_2 = Mass of object pulled towards the earth;
r = distance between the particles (radius of earth - 6,371 km);
G= universal gravitational constant, approximately 6.673×10^{-11} $m^3/(kg.s^2)$
F = force of attraction between two particles

$$F = G\frac{m_{Earth} \times m_2}{r^2} \Rightarrow F = \frac{G \times m_{Earth}}{r^2} \times m_2 \Rightarrow F = g \times m_2$$

Law of Universal Gravitation

Question:
Using the data provided on the previous slide, determine *g*, the value of acceleration due to the earth's gravitational pull.

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